

Department of Mathematics and Applied Mathematics

Module 3MS: Metric Spaces

Class test 1

23 March 2023

Time : 18:00 – 20:00

Total marks: 40

Please answer Questions 1 and 2 in one answer book, and Questions 3 to 5 in another.

1. (a) Consider $\mathbb{R}^2$ with the Euclidean metric d_E . Let $A = \{(a, b) \in \mathbb{R}^2 : 0 < a \leq 2095\}$.
 - i. Write down $\overline{A}$. (No proof required.)
 - ii. Write down $\text{bd}(A)$. (No proof required.)
 - iii. Write down $\text{int}(A)$. (No proof required.)
- (b) Consider $\mathbb{R}^2$ with the discrete metric. Let $B = \{(a, b) \in \mathbb{R}^2 : -10 < a < 1\}$.
 - i. Write down $\overline{B}$. (No proof required.)
 - ii. Write down $\text{bd}(B)$. (No proof required.)
 - iii. Write down $\text{int}(B)$. (No proof required.)

[3]

2. Let $X = \{x \in \mathbb{R} : x > 0\}$. For $x, y \in X$, define:

$$\delta(x, y) = \left| \frac{1}{x} - \frac{1}{y} \right|$$

- (a) Prove carefully that δ is a metric on X .
- (b) Find the closed ball $B[4; 1]$ in (X, δ) .
- (c) Let (x_n) be a sequence in X and $a \in X$. Prove that $x_n \rightarrow a$ in (X, δ) if and only if $x_n \rightarrow a$ in (X, d_E) , where d_E is the usual Euclidean metric.

[13]

Please change to a new answer book now.

3. Let A be a subset of a metric space X . Prove that $\text{bd}(A) = \emptyset$ if and only if A is both closed and open in X .
4. Read the following theorem and proof, and then answer the questions that follow.

Theorem For subset A of a metric space (X, d) , $\overline{A} = X$ if and only if A has non-empty intersection with every open ball in (X, d) .

Proof

($\implies$) Suppose that $B(x; r)$ is an open ball in (X, d) . (1)

There exists a sequence (a_n) such that $a_n \in A$ for all n and $a_n \rightarrow x$. (2)

Choose $N \in \mathbb{N}$ such that $n \geq N \implies d(a_n, x) < r$. (3)

Then $B(x; r) \cap A \neq \emptyset$. (4)

($\Leftarrow$) Suppose $x \in X$. (5)

For any natural number n , let $c_n \in A \cap B(x; \frac{1}{n})$. (6)

It follows from this that $x \in \overline{A}$. (7)

- (a) How do we know that the sequence (a_n) mentioned in line (2) exists?
- (b) How do we know that the number N mentioned in line (3) exists?
- (c) Justify the claim in line (4).
- (d) How do we know that the c_n mentioned in line (6) exist?
- (e) Explain how line (7) follows from the previous working.
- (f) Why does the statement in line (7) finish the proof?
- (g) Does there exist an open subset, G , other than $\mathbb{R}$ itself, such that $\overline{G} = \mathbb{R}$ in $(\mathbb{R}, d_E)$? If so, provide one; if not, explain why not. [14]

5. Let $\mathbb{Q}$ denote the rational numbers, and let d_E be the usual Euclidean metric on $\mathbb{Q} \times \mathbb{Q}$. Prove that, in $(\mathbb{Q} \times \mathbb{Q}, d_E)$, every open set can be expressed as a union of clopen sets. (Recall that a set is called *clopen* if it is closed and open.) [4]

THE END